Marginalising a Gaussian over θ

Notation. $x = \delta\omega_m \in \mathbb{R}$, $\quad y = \delta\theta \in \mathbb{R}^5$, $\quad a = F_{\omega\omega} \in \mathbb{R}$, $\quad b = F_{\theta\omega} \in \mathbb{R}^5$ (so $F_{\omega\theta} = b^T$), $A = F_{\theta\theta} \in \mathbb{R}^{5\times5}$, $A = A^T$, $A > 0$.

1. Start

$$-2\ln P(x,y) = a\,x^2 + 2x\,b^T y + y^T A y \tag{1}$$

$$P(x) = \int d^5y\ \exp\Big[-\tfrac{1}{2}\big(a\,x^2 + 2x\,b^T y + y^T A y\big)\Big] \tag{2}$$

$$= e^{-\frac{1}{2}ax^2}\int d^5y\ \exp\Big[-\tfrac{1}{2}\big(y^T A y + 2x\,b^T y\big)\Big] \tag{3}$$

2. Complete the square

Define

$$y_* \equiv -x\,A^{-1}b. \tag{4}$$

Expand:

$$(y-y_*)^T A(y-y_*) = y^T A y - y_*^T A y - y^T A y_* + y_*^T A y_* \tag{5}$$

$$= y^T A y - 2\,y_*^T A y + y_*^T A y_* \qquad (A = A^T) \tag{6}$$

Cross term:

$$y_*^T A\,y = -x\,b^T (A^{-1})^T A\,y = -x\,b^T A^{-1} A\,y = -x\,b^T y \tag{7}$$

$$\Rightarrow\ -2\,y_*^T A y = 2x\,b^T y \tag{8}$$

Constant term:

$$y_*^T A y_* = x^2\,b^T A^{-1} A\,A^{-1} b = x^2\,b^T A^{-1} b \tag{9}$$

Therefore

$$(y-y_*)^T A(y-y_*) = y^T A y + 2x\,b^T y + x^2\,b^T A^{-1} b \tag{10}$$

$$\Rightarrow\ y^T A y + 2x\,b^T y = (y-y_*)^T A(y-y_*) - x^2\,b^T A^{-1} b \tag{11}$$

3. Substitute back

$$P(x) = e^{-\frac{1}{2}ax^2}\,e^{+\frac{1}{2}x^2\,b^T A^{-1} b}\int d^5y\ \exp\Big[-\tfrac{1}{2}(y-y_*)^T A(y-y_*)\Big] \tag{12}$$

$$= \exp\Big[-\tfrac{1}{2}x^2\big(a - b^T A^{-1} b\big)\Big]\ I(x), \tag{13}$$

$$I(x) \equiv \int_{\mathbb{R}^5} d^5y\ \exp\Big[-\tfrac{1}{2}(y-y_*)^T A(y-y_*)\Big] \tag{14}$$

4. Shift the integration variable

x is fixed inside the integral, so y_* is a constant vector. Let $u = y - y_*$:

$$d^5u = d^5y \qquad (\text{Jacobian } = 1) \tag{15}$$

$$y \in (-\infty, \infty)^5 \;\Rightarrow\; u \in (-\infty, \infty)^5 \tag{16}$$

$$I(x) = \int_{\mathbb{R}^5} d^5u \; \exp\Big[-\tfrac{1}{2}\, u^T A u \Big] \tag{17}$$

No x appears on the right-hand side.

5. Evaluate the remaining integral

Diagonalise $A = O^T \Lambda O$, $O^T O = \mathbb{1}$, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_5)$, $\lambda_k > 0$. Let $v = Ou$, $|\det O| = 1$:

$$u^T A u = u^T O^T \Lambda O u = v^T \Lambda v = \sum_{k=1}^{5} \lambda_k v_k^2 \tag{18}$$

$$I = \int d^5v \; \exp\Big[-\tfrac{1}{2} \sum_{k=1}^{5} \lambda_k v_k^2 \Big] = \prod_{k=1}^{5} \int_{-\infty}^{\infty} dv_k \; e^{-\frac{1}{2}\lambda_k v_k^2} \tag{19}$$

$$= \prod_{k=1}^{5} \sqrt{\frac{2\pi}{\lambda_k}} = \frac{(2\pi)^{5/2}}{\sqrt{\lambda_1 \cdots \lambda_5}} = \frac{(2\pi)^{5/2}}{\sqrt{\det A}} \tag{20}$$

6. Result

$$P(x) = \frac{(2\pi)^{5/2}}{\sqrt{\det F_{\theta\theta}}} \; \exp\Big[-\tfrac{1}{2}\, x^2 \big(F_{\omega\omega} - F_{\omega\theta} F_{\theta\theta}^{-1} F_{\theta\omega} \big) \Big] \tag{21}$$

$$F_{\mathrm{marg}} = F_{\omega\omega} - F_{\omega\theta} F_{\theta\theta}^{-1} F_{\theta\omega} \tag{22}$$

$$\sigma^2(\omega_m) = \frac{1}{F_{\mathrm{marg}}} \tag{23}$$

7. Same thing as the inverse Fisher matrix

Block inverse of $F = \begin{pmatrix} a & b^T \\ b & A \end{pmatrix}$:

$$(F^{-1})_{\omega\omega} = \big(a - b^T A^{-1} b \big)^{-1} \tag{24}$$

$$\Rightarrow\; \sigma^2(\omega_m) = (F^{-1})_{\omega\omega} \tag{25}$$